# Statistical Signal Processing
# Lecture 09:
# Model-based (parametric) estimation of PSD

**Literature and Problems: See course schedule**

## Agenda (preliminary):

**Disadvantages for non-parametric estimation: DFT-based and correlation-based**

**Model-based estimation in general**

- **Observed signal assumed to be generated by LTI-filtering of white noise**
- **Method: estimate LTI-system (filter coefficients) → PSD estimate**
- **Signal models: AR, MA, ARMA and system orders**
- **SLIDE: Overview of methods for spectral estimation**

**AR(p), AutoRegressive models**

- **Most used model, best for spectra characterized by peaks**
- **AR(p) difference equation and LTI-filter transfer function H(z)**
- **Overall method: estimate $r_x(k)$, assume order p, find AR-coefficients from $r_x(k)$ estimate → estimated H(z) → PSD estimate**
- **Yule-Walker equations: connection between H(z) coefficients and $r_x(k)$**
- **Solution of Yule-Walker equations: YW-method, covariance method, modified covariance method and Burg method**
- **Influence of assumed model order p**
- **Guidelines for choosing p: plot estimated model error as function of p or use criteria from information theory: AIC, Akaike Information Criteria or MDL, Minimum Description Length (both penalizing high p)**
- **SLIDE: Model based spectral estimation using AR models**
- **DEMO: examples**

**MA(q), Moving Average models**

- **Best for broadband, smooth spectra**
- **MA(q) difference equation and LTI-filter transfer function H(z)**
- **YW equations becomes non-linear in unknown H(z) coefficients**
- **Simpler approach using the fact that $r_x(k) = 0$ for $|k| > q$: DTFT of windowed $r_x(k)$ estimate using rectangular window of width 2q+1**
- **SLIDE: Model based spectral estimation using MA models**
- **DEMO: examples**

**ARMA(p,q), AutoRegressive Moving Average models**

- **ARMA(p,q) difference equation and LTI-filter transfer function H(z)**
- **Complicated robust methods exists**
- **Simpler three-step approach using MYWE, Modified Yule-Walker Equations**
- **SLIDE: Model based spectral estimation using ARMA models**
- **DEMO: examples**